



Genome-wide identification of *CAMTA* gene family members in *Phaseolus vulgaris* L. and their expression profiling during salt stress

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Abstract

The calmodulin-binding transcriptional activator (CAMTA) family was first observed in tobacco (NtER1) during a screening for the CaM-binding proteins, which are known to be one of the fast response stress proteins. Due to the increased importance of plant transcription factors in recent years; genome-wide identification of *CAMTA* genes has been performed in several plant species, except for *Phaseolus vulgaris*. Therefore, our aim was to identify and characterize *CAMTA* genes in *P. vulgaris* via in silico genome-wide analysis approach. Our results showed a total of eight *CAMTA* genes that were identified and observed on five out of 11 chromosomes of *P. vulgaris*. Four gene couples were found to be segmentally-duplicated and these segmental duplication events were shown to occur from 29.97 to 92.06 MYA. The phylogenetic tree of CAMTA homologs from *P. vulgaris*, *A. thaliana*, and *G. max*. revealed three groups based on their homology and the intron numbers of *Pvul-CAMTA* genes, ranged from 11 to 12. According to the syteny analysis; *CAMTA* genes of *P. vulgaris* and *G. max* revealed higher similarity, because they have highly similar genomes compared to *A. thaliana*. All *Pvul-CAMTA* genes were targeted by miRNAs, which play a role in response mechanism of salt stress. To detect expression levels in different plant tissues, mRNA analysis of *Pvul-CAMTA* genes were performed using publicly available expression data in Phytozome v12.1. In addition, responses of *Pvul-CAMTA* genes to salt stress, were also examined via both RNAseq and qRT-PCR analysis. To identify and to obtain insight into biological functions of *CAMTA* genes in the genome of *P. vulgaris*, several analyses were conducted using many online and offline bioinformatic tools, genome databases and qRT-PCR analyses. Due to this study being the first in the identification of *CAMTA* genes in *P. vulgaris*, this study could be considered as an useful source for future *CAMTA* genes studies in either *P. vulgaris* or comparative different plant species.

Keywords *Phaseolus vulgaris* · CAMTA · RNAseq · qRT-PCR · Genome-wide identification

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Background

Plants are exposed to a variety of unsuitable environmental conditions, due to their sessile nature for most of the times. These unexpected environmental conditions are known to be caused by; ‘biotic’ and ‘abiotic’ stressors, and both cause deficiencies in development and productivity of plants, mostly through similar pathways or mechanisms [1]. To elucidate mechanisms underlying stress response in plants; have been one of the major topics of interest for plant scientists for many years. Even though many genes and pathways have been identified thus far, there are still many of them that require to be clarified, which might play a role in stress response and adaptation [2].

In recent years, studies conducted in plant transcription factors (TFs) have been rapidly increasing, due to the importance of TFs in regulation of stress related genes.

Following this development, identification of TFs in plants at a genome-wide level, has increased in popularity, as an important source for the studies that are aimed to generate stress conditions tolerant plants [1–3]. TFs are a group of main proteins that; interact with *cis* elements that are located in the upstream regions of the promoter, and they regulate which expression of genes transcription factors play an important role in many physiological processes such as; metabolic equilibrium, cell cycle progression, growth, development, and production. Transcription factors affecting the life cycles of organisms play a crucial role in cell and non-cell signaling [4].

Ca^{+2} ions play an important role in intracellular signaling and gene transcription, as a secondary messenger in eukaryotes. Many sensor proteins that are responsible for detecting intracellular Ca^{+2} changes in the cell against stress stimuli, have been discovered so far [5]. Among them; Calmodulin (CaM) is considered as one of the most studied protein, that have shown to play role in calcium signal transduction. The calmodulin-binding transcriptional activator (CAMTA) family was first observed in tobacco (NtER1) during a screening for the CaM-binding proteins, which are known as fast stress response elements [6, 7]. CAMTA protein domains include functional domains such as; (1) CG-1 DNA-binding domain containing NLS (nuclear localization signal) at the N-terminus, (2) Ankyrin repeats, responsible for mediating protein–protein interactions, and (3) CaMBD, as well as a varying number of IQ motifs (IQXXRGXXR), which bind with CaM in a Ca^{+2} - independent manner [8–12].

Several CAMTA genes/proteins have been discovered in many different plant species via in silico genome-wide identification approach, except for *Phaseolus vulgaris* (common bean) thus far [4, 5, 13, 14]. Due to the importance of common bean in the human diet and economies of developing

countries; this study primarily aimed to identify and characterize CAMTA genes in *P. vulgaris* and to explore the roles of CAMTA genes in stress responses. Salt stress was applied to two Turkish common bean cultivars named as; Zulbiye and Yakutiye which are known to be salt-susceptible and -tolerant, respectively. Along with many bioinformatic tools for the identification and characterization of CAMTA genes in common bean, RNAseq analysis and qRT-PCR technique were also applied for the investigation of the genes potential roles in salt stress response.

Results and discussion

Identification of *Pvul*-CAMTA genes in *Phaseolus vulgaris* genome

To date, six CAMTA genes from *Arabidopsis*, seven from *Populus trichocarpa*, six from *Gossypium arboreum*, seven from *Gossypium raimondii*, nine from *Gossypium hirsutum* [4], nine from *Zea mays* L. and 15 from *Glycine max* have been discovered and characterized [8, 13–15]. In this study, a total of 8 putative *Pvul*-CAMTA genes were identified in *P. vulgaris* using in-silico bioinformatic tools which were described in the materials and methods section. These putative genes were named; *Pvul*-CAMTA-1 to *Pvul*-CAMTA-8, in addition, some important information regarding these genes was provided in Table 1. The identified *Pvul*-CAMTA proteins were found to contain 911 (*Pvul*-CAMTA-7) to 1086 (*Pvul*-CAMTA-1) aminoacid residues, and these numbers were found similar to previously identified CAMTA proteins from different plants species (Table 1) [8, 13–15].

The molecular weight and theoretical isoelectric points of putative *Pvul*-CAMTA proteins ranged from; 102.77

Table 1 Information of eight *Pvul*-CAMTA proteins

Gene name	Transcript name	NCBI Locus ID	Chr. no.	Chr. location	Length (aa)	Mol. wt (kDa)	pI	Subcellular localization
<i>Pvul</i> -CAMTA-1	Phvul.002G209300.1	XP_007159108.1	2	36,924,678-36,935,363	1086	121.64	5.66	nucl: 12, cyto: 1
<i>Pvul</i> -CAMTA-2	Phvul.002G256500.1	XP_007159660.1	2	42,310,465-42,317,944	1105	125.26	5.57	nucl: 10, cyto: 2, vacu: 1
<i>Pvul</i> -CAMTA-3	Phvul.L002946.1	XP_007158673.1	2	32,095,502-32,104,715	987	109.53	5.68	nucl: 14
<i>Pvul</i> -CAMTA-4	Phvul.006G206400.1	XP_007148412.1	6	30,900,851-30,911,858	1076	121.66	5.82	nucl: 12, cyto: 1
<i>Pvul</i> -CAMTA-5	Phvul.003G119800.1	XP_007154445.1	3	29,859,348-29,867,209	997	112.12	7.43	nucl: 9, cyto: 3, chlo: 1
<i>Pvul</i> -CAMTA-6	Phvul.003G111900.1	XP_007154355.1	3	28,187,173-28,194,709	922	104.78	6.76	nucl: 11, cyto: 1, plas: 1
<i>Pvul</i> -CAMTA-7	Phvul.009G254500.1	XP_007138978.1	9	36,734,077-36,745,918	911	102.77	6.78	nucl: 9, cyto: 4
<i>Pvul</i> -CAMTA-8	Phvul.001G263000.1	XP_007163775.1	1	51,711,829-51,722,211	966	108.64	5.22	nucl: 13

(Pvul-CAMTA-7) to 125.26 (Pvul-CAMTA-2) kDa and 5.22 (Pvul-CAMTA-8) to 7.43 (Pvul-CAMTA-5), respectively. *Pvul-CAMTA* genes were found to be distributed on Chr-1, -2, -3, -6 and -9 among a total of 11 common bean chromosomes. The highest number (three genes) of *Pvul-CAMTA* genes were found on Chr-2 (Table 1; Fig. 1).

The chromosome locations of *Pvul-CAMTA* genes demonstrated that; four gene couples were segmentally-duplicated and these segmental duplication events were known to occur from 29.97 to 92.06 MYA (Table 2). In addition to this, Ka/Ks ratio which were less than '1' for all duplication events showed that the evolution of these gene pairs has occurred under the effects of purifying selection (Table 2) [5].

Phylogenetic analysis, gene structure, conserved motifs, subcellular localization and promoter analysis of *Pvul-CAMTAs*

A total of 31 CAMTA proteins from *P. vulgaris*, *A. thaliana* and *G. max* were divided into three groups, based on their homology (Group A, B and C). Group C composed of the largest group of proteins with; 14 CAMTA proteins. Whereas, Group A was the smallest with eight from three different plant species (Fig. 2). Pvul-CAMTA-3 and -8 were clustered into group A, together with AtCAMTA4 and 5 CAMTAs from *G. max*. Pvul-CAMTA-6 and Pvul-CAMTA-7 were classified in group B together with 2 CAMTAs from *A. thaliana* (AtCAMTA5 and AtCAMTA6) and 5 from *G. max*. Group C contained Pvul-CAMTA-1, Pvul-CAMTA-4 and Pvul-CAMTA-5 proteins together with AtCAMTA1,-2,-3 which were previously reported to play roles in stress responses in *A. thaliana*, and 7 CAMTA proteins from *G. max* (Fig. 2) [15–17]. In addition, the

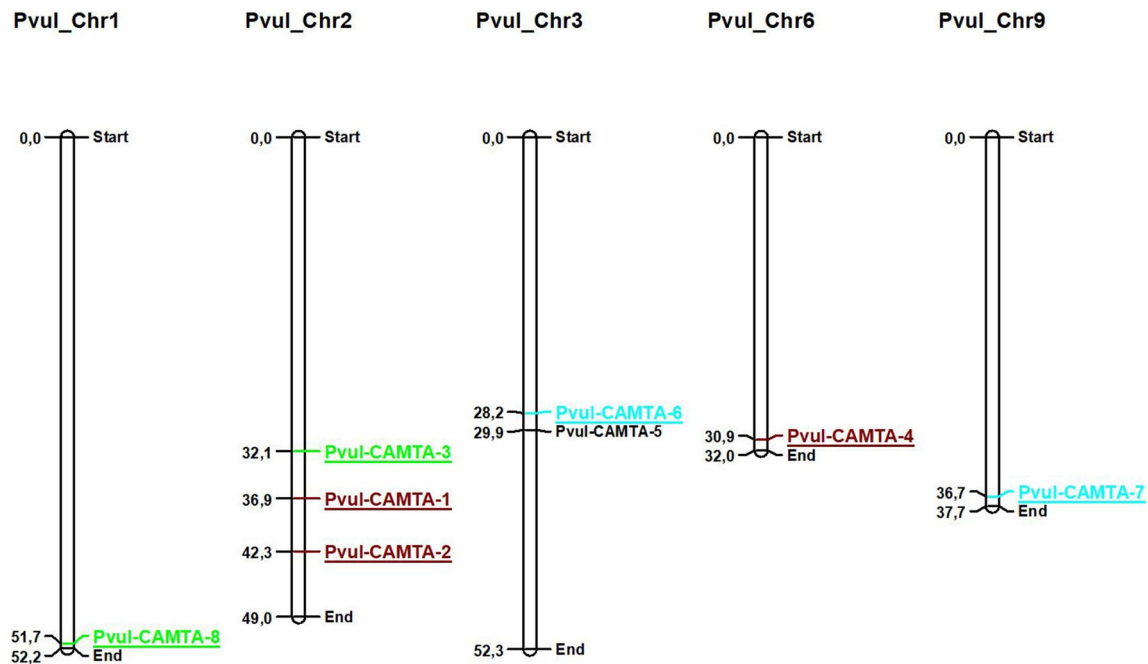
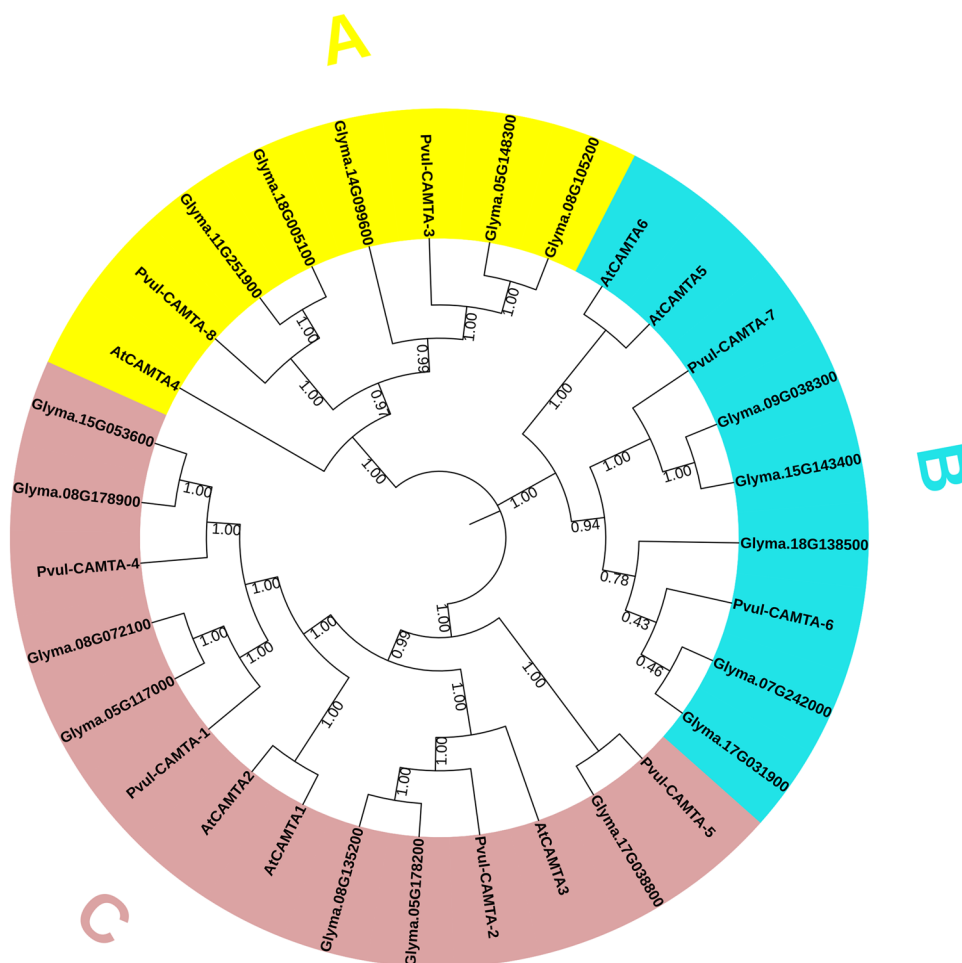


Fig. 1 Distribution of *Pvul-CAMTA* genes on chromosomes of *P. vulgaris*. The gene pairs with the same color indicate occurrence of duplication between them

Table 2 The Ka/Ks ratios and date of duplication for *CAMTA* genes in *P. vulgaris*

Duplicated CAMTA gene 1	Duplicated CAMTA gene 2	Ka	Ks	Ka/Ks	Date (mya) $T = Ks/2\lambda$	Selective pressure	Duplicate type
<i>Pvul-CAMTA-1</i>	<i>Pvul-CAMTA-4</i>	0.1922	0.5071	0.331	29.97	Purifying selection	Segmental
<i>Pvul-CAMTA-2</i>	<i>Pvul-CAMTA-4</i>	0.4616	1.5577	0.1848	92.06	Purifying selection	Segmental
<i>Pvul-CAMTA-6</i>	<i>Pvul-CAMTA-7</i>	0.205	0.5911	0.3089	34.93	Purifying selection	Segmental
<i>Pvul-CAMTA-8</i>	<i>Pvul-CAMTA-3</i>	0.2351	0.5405	0.4255	31.94	Purifying selection	Segmental

Fig. 2 Phylogenetic tree of CAMTA homologs from *P. vulgaris*, *A. thaliana* and *G. max*



uplicated CAMTA gene pairs (shown in Table 2) were grouped in the same clade tree as follows; *Pvul-CAMTA-1/Pvul-CAMTA-4* and *Pvul-CAMTA-2/Pvul-CAMTA-4* duplicated pairs were present in Clade C, *Pvul-CAMTA-6/Pvul-CAMTA-7* pair was present in Clade B and *Pvul-CAMTA-8/Pvul-CAMTA-3* was in Clade A (Fig. 2).

According to Fig. 3 which, reflects the genomic structure of *Pvul-CAMTA* genes; the intron numbers of *Pvul-CAMTA* genes ranged from 11 to 12 and are similar to the intron

numbers of CAMTA genes from; *A. thaliana*, *M. truncatula*, *G. max*, and *Z. mays*. Suggesting an important conservation of CAMTA genes across plant species [13–15] (Fig. 3).

Domain structure analysis of *Pvul-CAMTAs* exhibited that; *P. vulgaris* CAMTAs contained all conserved CAMTA domains including CG-1 DNA binding domain, ankyrin repeats, IQ motifs and IPT domain which is also known as TIG domain (Fig. 4). Some of the previous studies revealed that CAMTAs could be divided into two groups based on

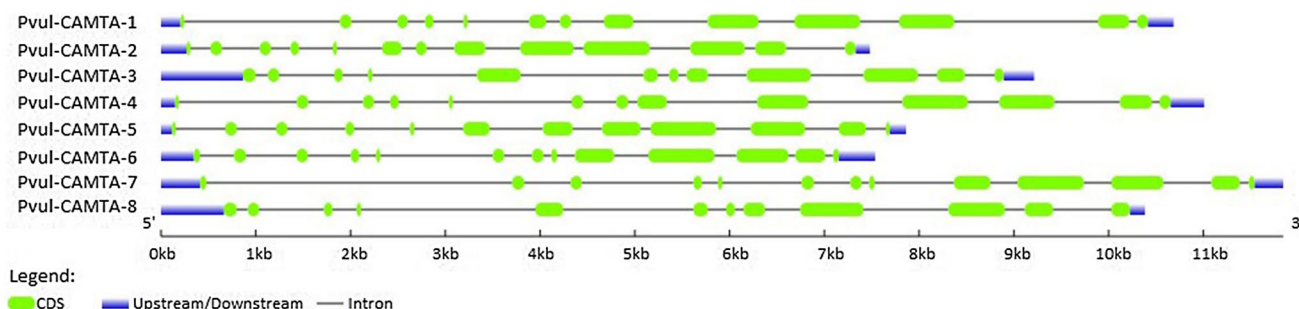


Fig. 3 Gene structure of CAMTAs in *P. vulgaris*

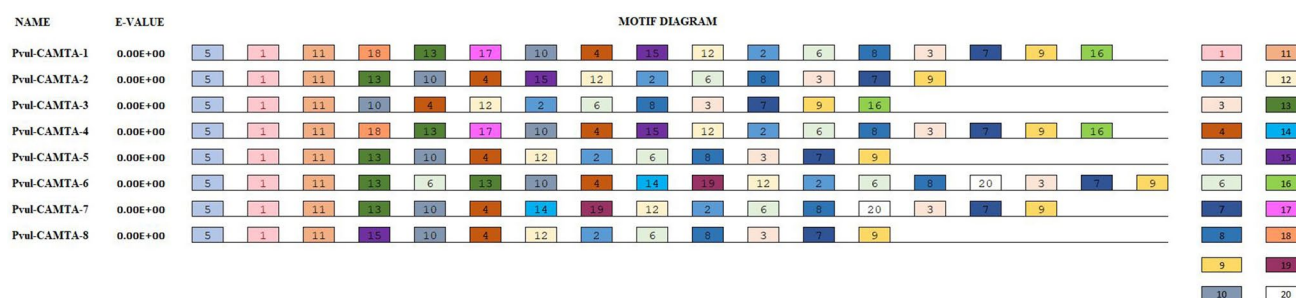


Fig. 4 Conserved motifs in CAMTA proteins identified using MEME suite program. The motifs designated with a specific color

whether the TIG domain is present or not [18]. According to this classification; *P. vulgaris* is part of the TIG contained CAMTA proteins category whereas, *A. thaliana* and *C. rubella* are the plants which have non-TIG CAMTA proteins [18].

Some important details of the conserved domains of PvuL-CAMTA proteins have been listed in Table 3.

The putative cis acting elements of *PvuL-CAMTA* genes (approximately – 2 kb upstream region), were detected with

the use of the promotor sequences obtained from Phytozome v12.1 database. A various number of cis acting elements were discovered, and are known to be related with development, environmental stress, hormone, light, promoter, and site binding.

MBS (MYB binding site involved in drought–inducibility) was observed in *PvuL-CAMTA*-2, -3, -4, -5, -6, -7 and -8 genes. It was reported that MBS element may be used to understand the molecular basis of both salt and

Table 3 Details of conserved motifs in CAMTA proteins

Motif	Name	E-value	Sites	Width (aa)	Domain
1	RKDGHNWRKKKDGKTVKEAHEKLVGNVEVLHCYYAHGED-NENFQRRSYW	5.20E–243	8	50	CG-1 DNA-binding domain
2	QGVHIAALGYDWAJNPIISAGVNIINFRDVNGWTALHWAAYF-GREKTVA	9.40E–191	8	50	Ankyrin repeat-containing domain superfamily
3	AAJRIQKKFRGWKGRKEFLMJRQRIKIQAHVRGHQVRKQYK	3.70E–157	8	42	IQ motif, EF-hand binding site
4	WACMFGEVEVP AEILQDGVLCQAPPHKGRVPFYVTCNSR-LACSEVREF	3.60E–161	8	50	IPT domain
5	JMZEAQHRWLPAEICAILRNYKKFKJTPEPPHRPPSG-SLFLDRKVLRY	4.90E–156	8	50	CG-1 DNA-binding domain
6	LVSAGASAGAVTDPTPQNPSGRTAADLASSNGHKJGAGYLSEVSLTSQL	1.10E–114	8	49	N/A
7	IIWSVGILEKVILRWRRKSGSLRGFRPET	1.80E–84	8	29	N/A
8	EDQQLKDSLAAVRNAAQAAARIQQAFRMHSFQRKQLA	5.00E–81	8	38	N/A
9	EGRKQAEERIEKALSRVKSVMRYPEARDQYRRL	5.40E–71	8	33	IQ motif, EF-hand binding site
10	DLSSLSDQLFSITDFSEWAYATETTKVJIVGSFLKSQ	6.60E–68	8	40	N/A
11	MLEPELEHIVLVHYREVKEGK	5.40E–47	8	21	CG-1 DNA-binding domain
12	DHLFERLLKDKLYEWLLEKITEGGKGTDV	2.80E–33	8	29	N/A
13	EGLKKLDSFGRWMTKEJGD	1.00E–13	6	19	N/A
14	EYRTPILHDPTAAMEEKYNWDEFRLQMRLAHLLF	2.50E–08	2	34	N/A
15	NVDFAEFFGSATEMLHLRLLEGFLSLKPVNPSNQVFEGDME-KRNLIKLI	9.50E–06	2	50	N/A
16	AVDGVEDWIDIDMLLDENFIPIAF	2.80E–04	2	25	N/A
17	NMQSSPGISWSTDECGDIDD	8.90E–04	2	21	N/A
18	YHMHFPFGDPGQFPIPGAIEYIPFVQEDKFRGNDTAYFEGEKAH-GIAPW	1.10E–03	2	48	N/A
19	VSPNRLKEARRFAFKTSFISKSWQYLKSIEDKQIPF	2.20E–03	2	37	N/A
20	PEDZARKIVAAMKIQHAFR	1.20E–02	2	19	N/A

drought stresses [19]. Two (*Pvul*-CAMTA-6 and -7) and four (*Pvul*-CAMTA-1, -4, -6 and -7) out of eight *Pvul*-CAMTA genes were found to have GT-1 and TGACG motifs- respectively, and these motifs are known to be salt stress responsive elements [1].

AREB-responsive elements (ABRE motif) were discovered to be included in *Pvul*-CAMTA-1, -5 and -8 genes. Some other salt stress related elements such as; G-box, and TGA-element were additionally identified. The presence of G-Box element was confirmed in all of the *Pvul*-CAMTA genes except for, *Pvul*-CAMTA-2, -3 and -7 and a TGA-element was discovered in all *Pvul*-CAMTA genes except in *Pvul*-CAMTA-8 (Supplementary Table 1). Wei et al. [5] indicated that some members of PtCAMTA (*Populus trichocarpa*) gene family contain ABRE and MBS cis acting elements [5]. In a previous study conducted by Buyuk et al. [1], GT-1 and TGACG motifs were identified as salt stress responsive elements [1]. Saeediazar et al. (2014) also indicated that some genes which contained cis acting elements such as; ABRE, G-box, MBS and TGA-element had regulatory roles in salt stress conditions. All of those stress related cis acting elements were detected in *Pvul*-CAMTA genes in this current study.

The subcellular localizations were predicted with WolfPSORT, and according to the results; *Pvul*-CAMTA genes were located in nucleous (nucl), cytosol (cyto), vacuolar membrane (vacu), chloroplast (Chlo), and plasma

membrane (plas) (Table 1). In recent studies, the CAMTA gene family was observed to be located in nucleous [4, 5].

Comparative and syteny events among CAMTA genes of *P. vulgaris*, *A. thaliana* and *G. max*

To calculate evolutionary relationships of the CAMTA gene family between *P. vulgaris*, *A. thaliana* and *G. max*; the syteny events among eight CAMTA proteins from *P. vulgaris*, six from *A. thaliana*, and 14 from *G. max* were analyzed. CAMTA genes of *P. vulgaris* and *G. max* revealed higher similarity due to the high similarity of the genomes (Fig. 5) [20].

Detection of miRNAs targeting *Pvul*-CAMTA genes

According to the results of psRNATarget server; the maximum expectation and the accessibility of messenger RNA (mRNA) target site to miRNA ranged from 2.0 to 3.0 and 9.859–24.695- respectively (Supplementary Table 2). Eight *Pvul*-CAMTA genes were targeted by miRNAs in 64 plant species. A total of 558 miRNA families were detected to target the eight *Pvul*-CAMTA genes (Supplementary Table 2). In *Arabidopsis* and *P. trichocarpa* plants exposed to salt stress; the miR398, miR482.2 and miR1450 were down-regulated whereas, miR156, miR158, miR159, miR165, miR167, miR168, miR169, miR171 family, miR319, miR393, miR394, miR396, miR397, miR530a, miR1445,

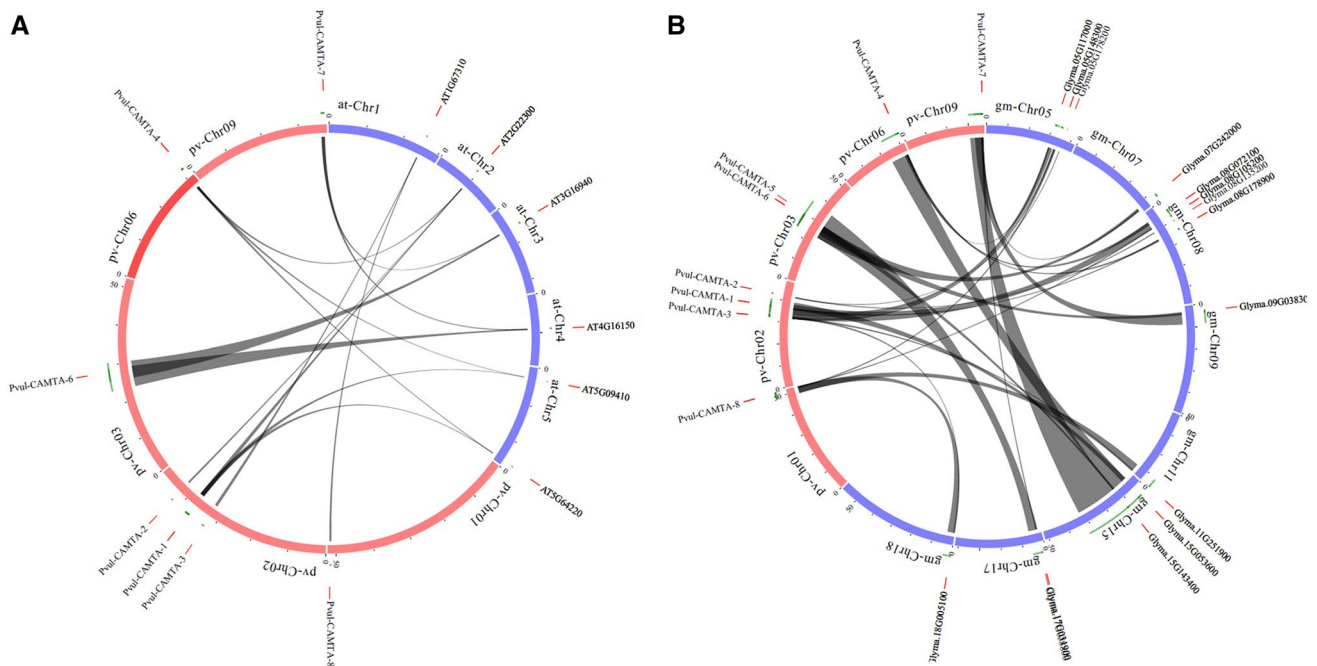


Fig. 5 Syteny events among CAMTA genes of *P. vulgaris*, *A. thaliana* and *G. max*. **a** Comparison of *P. vulgaris* with *A. thaliana*, **b** Comparison of *P. vulgaris* with *G. max*. (*P. vulgaris*, *A. thaliana* and *G. max* chromosomes were arranged in a circle)

miR1446 family, miR1447 were up-regulated [21, 22]. Similarly, miR159 in *Phaseolus vulgaris* as well as miR529 and miR5021 in bread wheat, were found to be up-regulated under salt stress conditions [23–25].

Tissue-specific mRNA levels of *Pvul-CAMTA* genes

In this study, we additionally aimed to perform mRNA analysis of *Pvul-CAMTA* genes using publicly available expression data in Phytozome v12.1 online plant genomics resource (<https://phytozome.jgi.doe.gov>). The heatmap, which was derived from mRNA levels, shows the expression variance of identified eight *Pvul-CAMTA* genes in

different plant tissues. Which in turn helped us understand the roles of *Pvul-CAMTA* genes in different parts of common bean (Fig. 6). As seen in the Fig. 6, *Pvul-CAMTA*-2, -5 and -6 genes revealed relatively low expression levels in almost all tissues including; flower buds, flowers, leaves, stem 10, young pods, stem 19, young trifoliates, root 10, root 19, green mature pods and nodules. The rest of the *Pvul-CAMTA* genes exhibited relatively high mRNA levels in most of the developmental stages of common bean. *Pvul-CAMTA*-1 and -8, were the genes which exhibited the highest expression levels in almost all tissue types of common bean. In addition, *Pvul-CAMTA*-3, -4, -6 and -7 genes were also highly expressed in most of the tissue types.

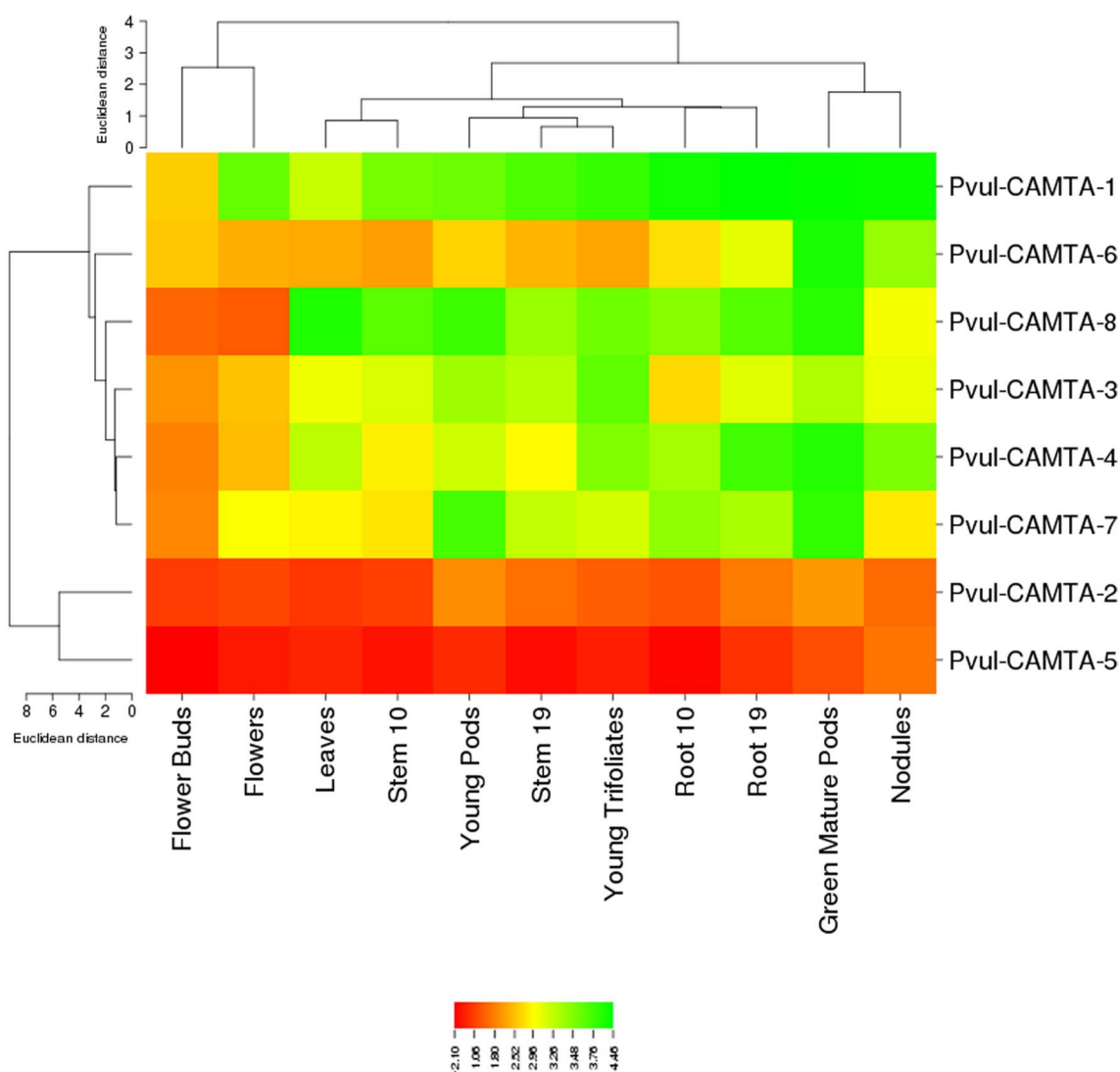


Fig. 6 Expression based-heat map of *Pvul-CAMTA* genes obtained from publicly available expression data in Phytozome v12.1. *FPKM (expected number of fragments per kilobase of transcript sequence per millions base pairs sequenced) units were used for the in-silico

expression levels. FPKM values were log2 transformed and the heatmap was produced with the algorithm CIMMiner (<http://discover.nci.nih.gov/cimminer>)

Roles of *Pvul-CAMTA* genes in response to salt stress

In this study, a comprehensive RNAseq analysis of *Pvul-CAMTA* genes were performed via publicly available SRA data that was reported by Hiz et al. [40]. Both leaf and root tissues were found to reveal differentially expression of some *Pvul-CAMTA* genes under salt stress conditions. The expression levels of *Pvul-CAMTA-1*, -2, -3, -4, -5 and -8 genes increased in salt treated leaf tissues compared to, non-treated leaf tissues whereas, *Pvul-CAMTA-6* gene showed same expression levels in both cases (Fig. 7). The only gene that showed decreased level of expression in salt treated leaf tissues compared to non-treated ones was, *Pvul-CAMTA-7*. In root tissues, half of the *Pvul-CAMTA* genes did not respond to salt stress. Only three of *Pvul-CAMTA* (*Pvul-CAMTA-1*, *Pvul-CAMTA-2* and *Pvul-CAMTA-4* genes) showed a difference in terms of expression level between salt-treated and non-treated control samples.

We performed qPCR assay to, explore the possible role of *Pvul-CAMTA* genes in 150 mM salt stress for a period of 9 days and the results were plotted on graph represented in this article as Fig. 8. Two common bean cultivars were used for the qPCR experiments, to reflect possible gene expression differences of *Pvul-CAMTA* in tolerant (Yakutiye) and sensitive (Zulbiye) cultivars. The results of qPCR reactions showed that; all *Pvul-CAMTA* genes except for *Pvul-CAMTA-7*, were highly expressed in the leaves of Yakutiye compared to Zulbiye in a statistically significant fashion (Fig. 8). In 2009, a previous study in *Arabidopsis* showed the role of CAMTA proteins in cold acclimation, and partly in salt response. In a similar fashion, we found out that *Pvul-CAMTA* genes are differentially expressed between -tolerant (Yakutiye) and -sensitive (Zulbiye) lines suggesting that, their most likely role is in stress response against salt stress. As previously indicated, Yakutiye is a tolerant cultivar to both NaCl and drought and in line with this, expression level of most *Pvul-CAMTA* genes were found to be higher than sensitive cultivar Zulbiye in a statistically significant manner.

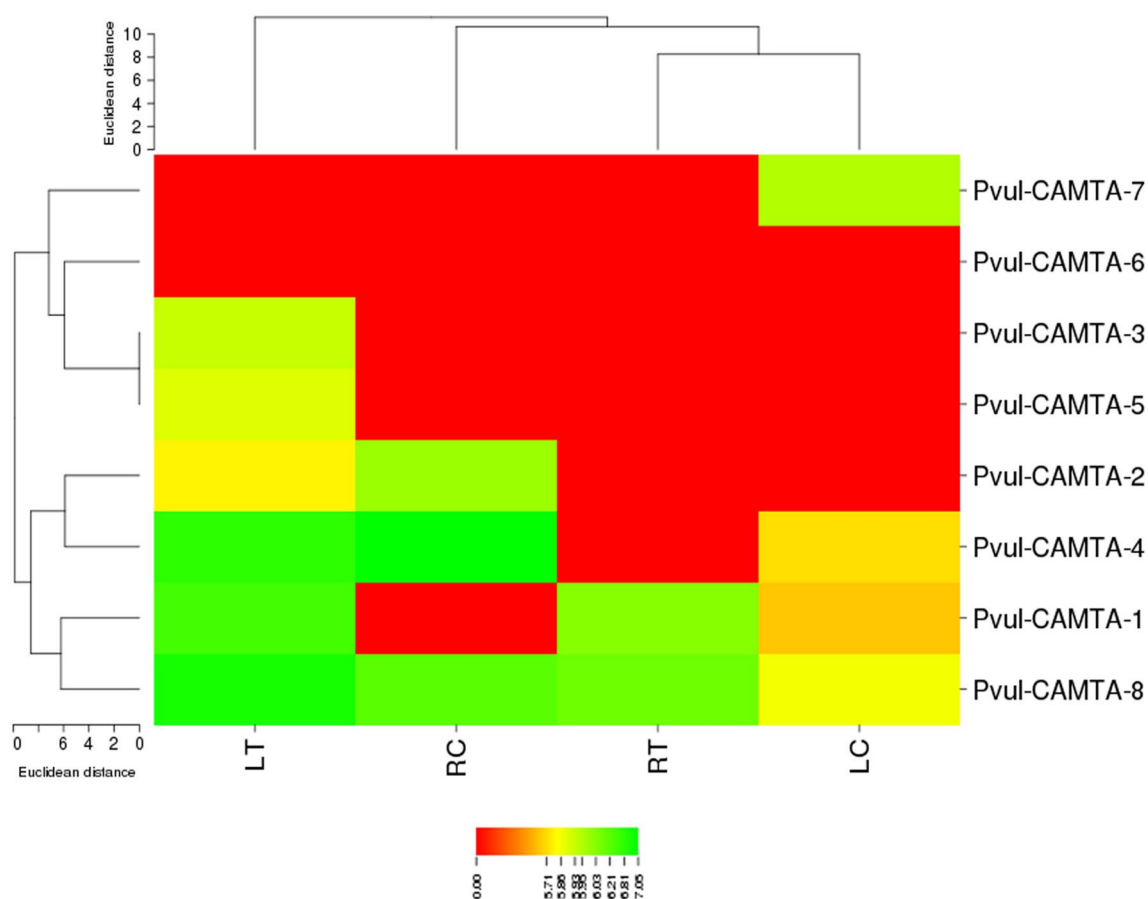


Fig. 7 Heat map of differentially expressed *Pvul-CAMTA* genes under normal and salt stress conditions derived from RNAseq analysis

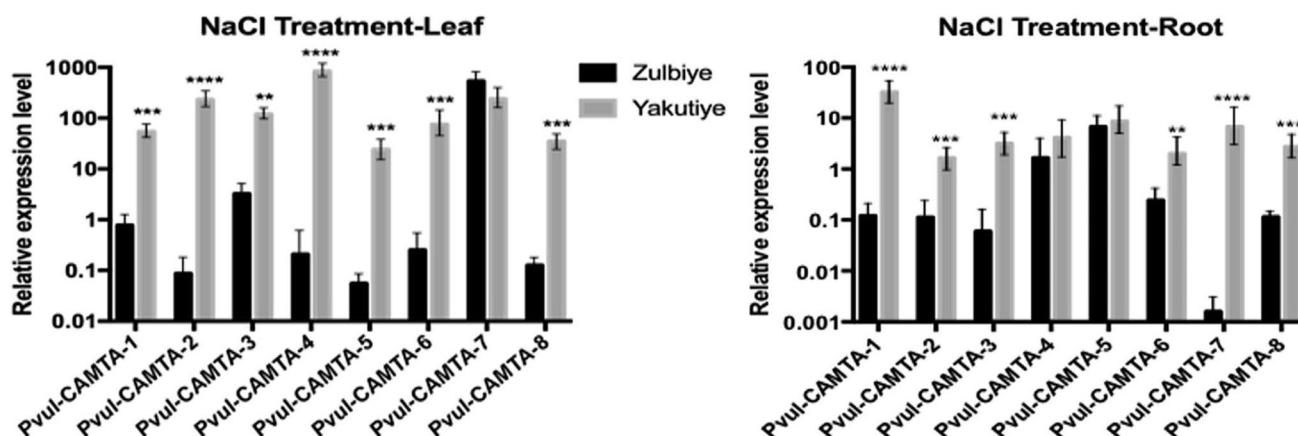


Fig. 8 Expression analysis of *PvuL-CAMTA* genes under NaCl stress. Errors bars represent the standart errors of three biological replicates and asterisks indicate genes that showed a significant difference in

expression level between Zulbiye and Yakutiye cultivars (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$)

Conclusions

To sum up, the study herein reports a comprehensive genome-wide identification of *CAMTA* genes in *P. vulgaris*. A total of 8 *CAMTA* genes were identified, and were given the names; *PvuL-CAMTA-1* to *PvuL-CAMTA-8*. To get insight into their biological functions in the genome of *P. vulgaris*, several analyses were conducted using many online and offline bioinformatic tools, genome databases, and qRT-PCR analyses. Due to this study being the first in the identification of *CAMTA* genes in *P. vulgaris*, this study could be considered as a useful source for the future studies concerning *CAMTA* genes either in *P. vulgaris* or comparative different plant species.

Methods

Identification of *CAMTA* proteins in *Phaseolus vulgaris* genome

Sequences of the *CAMTA* gene family of *Phaseolus vulgaris* were obtained from Phytozome database v12.1 (<http://www.phytozome.net>) using key search with Pfam Accession Number (PF03859, PF01833, PF12796, PF00612), and downloaded from Pfam Database (<http://pfam.xfam.org/>). To identify all the putative *CAMTA* proteins in *P. vulgaris*, both blastp at Phytozome database v12.1 (<http://www.phytozome.net>) and hidden Markov model (HMM), (<http://www.ebi.ac.uk>) searches were performed against the *P. vulgaris* genome [26]. Putative *P. vulgaris* *CAMTA* proteins were used for query in blastp (NCBI) for characterization of hypothetical proteins. Redundant sequences were removed by using the decrease redundancy tool (http://web.expasy.org/decrease_redundancy/) and sequences were checked

for *CAMTA* domains by using HMMER (<http://www.ebi.ac.uk>) and Pfam databases. The physicochemical properties of *CAMTA* proteins were calculated using ProtParam Tool (<http://web.expasy.org/protparam>).

Structure and physical locations of *CAMTA* genes and conserved motifs

The Gene Structure Display Server v2.0 (GSDS, <http://gsds.cbi.pku.edu.cn/>) was used [27] to obtain information on the exon – intron of *PvuL-CAMTA* proteins. Both genome sequences and the coding sequences were utilized for, predicting the positional information of the *PvuL-CAMTA* genes using Phytozome database v12.1, chromosomal locations. The sizes (bp) and intron numbers of *PvuL-CAMTA* genes were identified. The *PvuL-CAMTA* genes were plotted on all *P. vulgaris* chromosomes and pictured with MapChart [28]. To identify additional conserved motifs of *PvuL-CAMTA* proteins, we used the Multiple EM for Motif Elicitation tool (MEME v4.11.1; <http://meme-suite.org/>) [29].

Phylogenetic analysis and sequence alignment

The multiple sequence alignment of *PvuL-CAMTA* proteins was carried out using ClustalW [30]. The neighbor-joining (NJ) method with bootstrap value of 1000 replicates (MEGA7) was used for constructing phylogenetic trees and the tree was designed using Interactive tree of life (iTOL; <http://itol.embl.de/index.shtml>) [31].

Subcellular localization and promoter analysis of *PvuL-CAMTA* family

The 5' upstream regions (2 kb DNA sequence of each gene of the *PvuL-CAMTA* family) were obtained from Phytozome

database v11 and analyzed using the plantCARE database (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) for a cis-element scan. Subcellular Localization of the Pvul-CAMTA proteins was determined using WoLF PSORT (http://www.genscript.com/psort/wolf_psort.html) [32] and TargetP 1.1 (<http://www.cbs.dtu.dk/services/TargetP/>) [33].

In silico prediction of miRNA targets in *Pvul-CAMTA* genes

All known plant miRNA sequences were obtained from miRBase v21.0 (<http://www.mirbase.org>). psRNA Target Server (<http://plantgrn.noble.org/psRNATarget>) with default parameters used for the prediction of miRNAs, accordingly [34]. In-silico predicted miRNA targets were identified by BLASTX with $\leq 1e-10$ against common bean Expressed Sequenced Tags (ESTs) in the NCBI database.

Detection of gene duplication events and prediction of synonymous and non-synonymous substitution rates

Segmental duplicate gene pairs were examined on the Plant Genome Duplication Database server (<http://chibba.agtec.uga.edu/duplication/index/locus>), with a display range of 100 kb. CLUSTALW software was used for the prediction of amino-acid sequences of segmentally duplicated *Pvul-CAMTA* genes. The CODEML program in PAML (PAL2NAL) (<http://www.bork.embl.de/pal2nal>) was used to estimate the synonymous (Ks) and non-synonymous (Ka) substitution rates [35]. Time of duplication (million years ago, Mya) and divergence of each *Pvul-CAMTA* genes was determined using the following formula: $T = Ks/2\lambda$ ($\lambda = 6.56E-9$) [36, 37].

Syntenic analysis

P. vulgaris and *A. thaliana*, *P. vulgaris* and *G. max* of orthologue CAMTA genes were identified with Plant Genome Duplication Database (PGDD; <http://chibba.agtec.uga.edu/duplication/>) [38]. Then, the protein sequences of orthologue were retrieved from Phytozome v12.1. The obtaining syntenic map was created using iTAK - Plant Transcription factor & Protein Kinase Identifier and Classifier (<http://itak.feilab.net/cgi-bin/itak/index.cgi>) [39].

Gene expression analysis in silico

The expression levels of *Pvul-CAMTA* genes examined in special tissue libraries of plants at different stages of growth, including root_10, nodules, root_19, young pods, stem_10, stem_19, green mature buds, leaves, young trifoliate, flower buds, and flowers were retrieved from Phytozome

Database v12.1 (http://phytozome.jgi.doe.gov/pz/porta1.html#!info?alias=Org_Pvulgaris). FPKM (expected number of fragments per kilobase of transcript sequence per millions base pairs sequenced) units were used for the expression levels in silico. FPKM values were log2 transformed and the heatmap was produced with the algorithm CIM-Miner (<http://discover.nci.nih.gov/cimminer>).

Identified expression level of *Pvul-CAMTA* genes through transcriptome data

Illumina RNA-seq data was obtained from Sequence Read Archive (SRA) in order to measure the expression levels of *Pvul-CAMTA* genes. The accession numbers SRR957667 (control leaf), SRR958472 (salt-treated root), SRR958469 (control root), and SRR957668 (salt-treated leaf) were used for this purpose [40]. The analysis of transcriptomic data was performed as described by Buyuk et al. [1] using log2 transformed data. Finally, the heat maps of hierarchical clustering were constructed using the PermutMatrix [41].

Plant materials, growth conditions, and stress treatments

The seeds of salt tolerant bean cv. ‘Yakutiye’ and salt sensitive bean cv. ‘Zulbiye’ were supplied from ‘Transitional Zone Agricultural Research Institute, Eskişehir, Turkey’ [42]. According to our previous studies, ‘Yakutiye’ was declared as salt tolerant while ‘Zulbiye’ was declared as salt sensitive [1–3]. For gene expression analysis, seeds of both cultivars were germinated, following the surface sterilization in a solution containing 5% (v/v) hypochlorite for 5 min, and were grown hydroponically in pots containing 0.2L of modified 1/10 Hoagland’s solution. Hoagland solution includes macronutrients (K_2SO_4 , KH_2PO_4 , $MgSO_4 \cdot 7H_2O$, $Ca(NO_3)_2 \cdot 4H_2O$ and KCl) and micronutrients (H_3BO_3 , $MnSO_4$, $CuSO_4 \cdot 5H_2O$, NH_4Mo , $ZnSO_4 \cdot 7H_2O$) with a final concentration of ions as 2 mM Ca, 10^{-6} M Mn, 4 mM NO_3 , $2 \cdot 10^{-7}$ M Cu, 1 mM Mg, 10^{-8} M NH_4 , 2 mM K, 10^{-6} M Zn, 0.2 mM P, 10^{-4} M Fe and 10^{-6} M B [43]. Common bean seedlings were incubated in a controlled environmental growth chamber under light with $250 \text{ mmol m}^{-2} \text{ s}^{-1}$ photosynthetic photon flux at 25 °C and 70% relative humidity. Salt stress was applied with Hoaglands solution as well as a 150 mM NaCl (for moderate salinity stress) for a period of 9 days, after common bean seedlings reached the first trifoliate stage in the growth chamber. Following the ninth day of stress application, root and leaf tissues of two different common bean cultivars were sampled, and submitted to qPCR analysis. Three biological replicates were used for the qPCR reactions.

RNA isolation, cDNA synthesis and quantitative real-time PCR analysis

Total RNAs of leaf and root tissues of the two bean cultivars, were extracted using NucleoSpin® RNA Kit (Macherey–Nagel, Germany) in accordance with the manufacturer's protocol. NanoDrop Lite UV–vis spectrophotometer was used to determine RNAs quality/quantity additionally, the presence of RNAs were also confirmed using gel electrophoresis, comprised of; 1.5% agarose (NanoDrop Technologies, Wilmington, DE, USA). cDNA synthesis was performed using 2 µg of RNA with a high fidelity cDNA synthesis kit (Roche), which contains 2.5 µM Anchored-oligo (dT)₁₈, 1X transcriptor high fidelity reverse transcriptase reaction buffer, 20U protector RNase inhibitor, 1 mM deoxynucleotide mix, 5 mM DTT and 10U transcriptor high fidelity reverse transcriptase at final concentration. The following incubation conditions were applied; 10 min at 65 °C, 30 min at 55 °C and 5 min at 85 °C. cDNA quantity/quality was also measured using a Nanodrop ND- Spectrophotometer Lite.

qPCR was performed with Light Cycler® Nano System (Roche), thermal cycler. The primer sequences of the target genes and housekeeping gene, which are used for normalization, were designed with Primer3 program based on the sequences of eight predicted *Pvul*-CAMTAs (Supplementary Table 2). All qPCR reactions were performed in three independent biological and technical triplicates, with a template free control to check for any contaminations. Amplifications of PCR product were monitored using an intercalator-based method including; SYBR Green I dye. After pre-denaturation for about 10 min at 95 °C, 45 cycles of 15 s at 95 °C, 20 s at 60 °C and 20 s at 72 °C were applied. Melting-Curve analysis was performed to confirm the presence of a single product and the absence of primer-dimers.

The abundance of target gene transcripts was normalized to ACT and set relative to control plants according to the $2^{-\Delta\Delta CT}$ method [44] and the log2 transformed data was used for the given graphs. Changes in relative expression levels (REL) of the gene were checked for statistical significance in accordance to the one-way ANOVA. Fisher's least significant difference test at 0.05 significant levels was performed.

Authors' contributions All authors contributed equally to this manuscript.

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Data availability All data generated or analysed during this study are included in this published article [and its supplementary information files]

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